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V-28 – Blend systems

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1.0 Purpose

The purpose of this bulletin is to define acceptable blending practices and to provide a guiding policy for blending systems.

2.0 Scope

This bulletin applies to all metering systems that blend two or more liquids for trade purposes.

3.0 Terminology

Additive Injector

An accessory item used in conjunction with a metering system to inject relatively small amounts of additives (i.e., less than 1%) into a primary product. Such additives alter the quality or characteristics of a product. In general, it is the same type of equipment as a dye injector.

Dye Injector

A component used in conjunction with a metering system to inject relatively small amounts (i.e., less than 1%) of dye into a primary product. Such dyes are added to identify a product, typically for excise purposes.

Ratio Blending

A process whereby two or more products are measured individually by a dedicated trade meter and then combined prior to delivery into the recipient vessel.

Sequential Blending

A process whereby a single trade meter is used to measure two or more products. In these systems, each product is measured individually, in sequence, and blending takes place in the recipient vessel.

Side stream Blending

A process whereby one or more products are metered into a line containing the primary product. The resulting blend is then measured as a blend by an approved and inspected meter, for trade purposes.

4.0 Background

The rise in popularity and, in some cases, legislated requirement for the use of alternative fuels such as ethanol and biodiesel, has led to a number of facilities being constructed that incorporate some manner of blending. Given the nature of the products and the need for accurate measurement and temperature conversion, it became evident that a guiding policy for such systems was required.

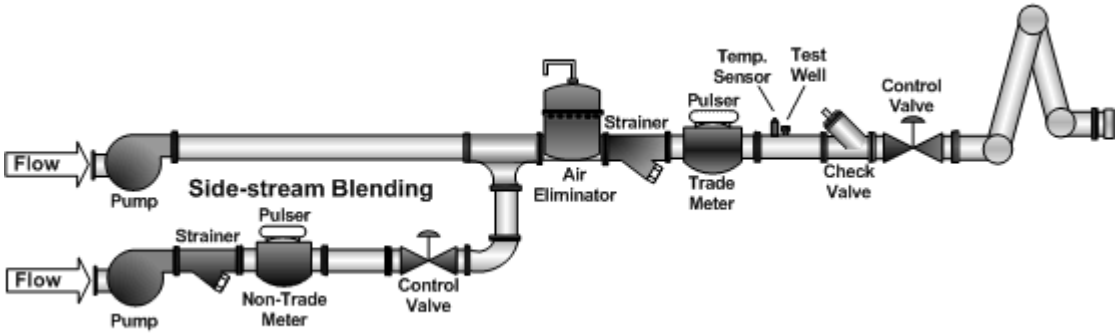
5.0 Policy

The following methods of blending or injecting products are recognized as acceptable, provided all the conditions noted below are met. Any other methods of blending or injecting products not already recognized in this bulletin will be evaluated on a case-by-case basis.

5.1 Side stream blending

> ℹ **Note:** The following metering systems represent minimum requirements for trade and general location of the system components. For reasons of safety and/or preference, many other components may be included. For the sake of simplicity,such components are not illustrated.

Side stream blending



▶ Description of Figure 1

As defined, a side stream blending system consists of one or more non-trade meters upstream of the final trade meter, which measures the low proportion component(s) into the majority component, prior to the final measurement. In these systems, the side-stream meters do not require approval or inspection as they are only used to establish the blend ratio. The final meter in the system, which measures the blended product, must be approved and inspected prior to use in trade. It may be equipped with an automatic temperature compensator (ATC). Such a device must base its conversion on the gross (uncompensated) volume derived from the trade meter and the temperature measured in that meter. No conversions or alterations to the values measured by the blended product (trade) meter may be made using values from side-stream meters.

The required information must be printed on conclusion of a trade transaction in accordance with Measurement Canada's Bulletin V-20 — Printer Requirements for Volumetric Liquid Meters Equipped with Automatic Temperature Compensation where two or more components of the product are combined prior to being measured by the trade meter. The information stated should be appropriate for the product blend only (i.e., gross, net, product name, etc.). For the selection of appropriate density

values, refer to Measurement Canada's Bulletin V-18 — Selection of Volume Correction Factor Tables and Standard Density Values for Some Common Products.

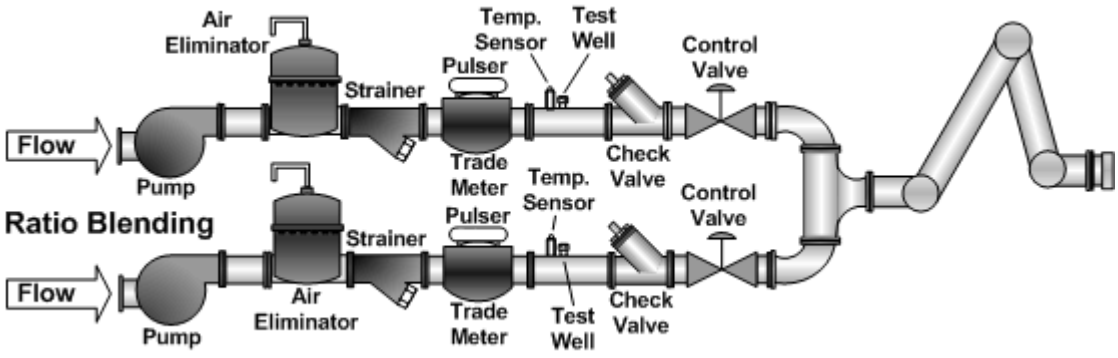
Additionally, the blending process may be proportional or non-proportional. In proportional systems, the blend ratio is established and maintained throughout the entire measurement transaction, which results in the correct ratio throughout the loading process. In non-proportional systems, the low proportion component is metered into the stream at the maximum rate of the side-stream meter. Once the correct amount of low proportion product has been reached, the side-stream flow is stopped. If the transaction was stopped short for any reason the blend ratio in the recipient vessel will be in error.

5.2 Ratio blending

In ratio blending, two or more trade meters simultaneously measure the component products to be blended prior to the actual blending process. In this case, all meters involved must be approved and inspected prior to their use in trade.

Each meter may be equipped with an ATC; however, if one meter is equipped with an ATC, all meters must be so equipped. Such ATCs must base their conversion on the gross (uncompensated) volume and temperature derived from the component meters. For inspection purposes, for systems other than retail motor fuel dispensers, the gross and net (compensated) indications of the measured values must be available for the individual meters. These values may be added to attain a total quantity delivered. Furthermore, the individual quantities may be declared and used for trade purposes.

Ratio blending



▶ Description of Ratio Blending

The information required to be printed is set out in Measurement Canada's Bulletin V-20, "Printer Requirements for Volumetric Liquid Meters Equipped with Automatic Temperature Compensation". For systems other than retail motor fuel dispensers, where two or more components of the product are measured separately, each through an approved meter, the information for each component may be stated (i.e., gross - component 1, gross - component 2, net - component 1, net - component 2 and the total net value which will be a summation of the net component products for the combined product delivered, etc.). When the system can provide the individual component information noted above for inspection purposes and for retail motor fuel dispensers, the printed ticket may state the required information for the blended product only (i.e., gross - blended product, net - blended product, density - blended product).

The blending process may be either proportional or non-proportional, but in either case, all meters involved in the process must be operated within their approved flow rate, with the exception of a short period at the start and finish of the transaction.

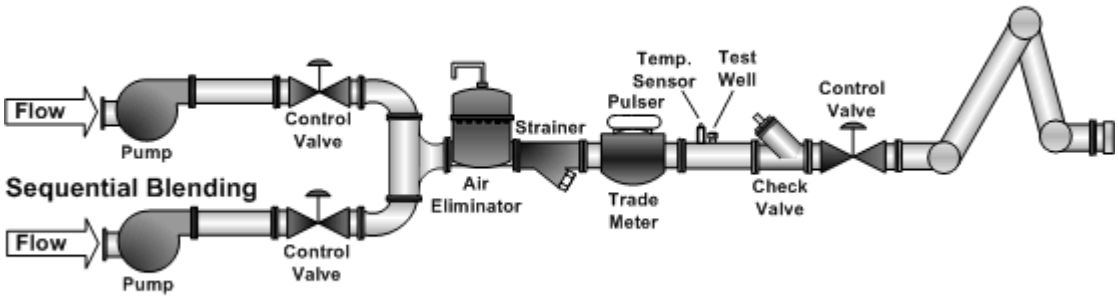
For the purpose of inspection, ratio blending systems must have the ability to test each meter separately.

5.3 Sequential blending

In the case of sequential blending, the component products are measured individually by a single trade meter.

The meter must be approved and inspected prior to use in trade. If there is a difference in the viscosity of the components, the register must employ a separate and specific meter factor for each component.

Sequential blending



▶ Description of Sequential Blending

These meters may employ an ATC, provided its conversions are based on the gross volume and temperature derived from the meter and the appropriate tables are employed for each product. At the time of inspection, the meter and ATC functions must be verified for each product the meter is intended to measure. Where a sequential blending system offers combinations of two or more grades of the same basic product (premium and regular gasoline) in addition to offering 100% deliveries of the components, it is recommended that the premium grade product be delivered last, to prevent the potential quality degradation of the next

delivery. The information required to be printed will be the same as the ratio blending system's printing requirements previously discussed in section 5.2.

5.4 Provisions for testing

Unless otherwise stated in the Notice of Approval, a ratio blending system shall be designed so that each trade meter may be tested individually, with the component product measured by the particular meter under test. A sequential blending system shall be designed so that the trade meter may be tested with each of the component products separately. The trade meter in a side stream blending system may be tested with either pure product or a blend of products that it commonly measures. Means are to be incorporated to allow the meters to be tested at their normal operating (preset) flow rate.

5.5 Additive injectors

Additive injectors are typically accessory items added to already approved metering systems. Such items do not require separate approval or inspection. Where their installation requires modification to the host device (physical or software), the approval holder for the host device must request a Modification Acceptance Letter (MAL) from the Volumetric Laboratory, which details the modifications required and declares the compatibly of the accessory, including the compatibility of the liquid being injected with the metering element. Additive injectors may be installed upstream or downstream of the metering element.

5.6 Dye injectors

Like additive injectors, dye injectors are also accessory items added to already approved metering systems. They also do not require separate approval or inspection. Where their installation requires modification of the host device (physical or software), the approval holder for the host device must request a MAL from the Volumetric Laboratory, which details the modifications required and declares the compatibility of the accessory. Dye injectors may be installed upstream or downstream of the metering element. To facilitate inspection, without degrading the product, dye injectors must be equipped with a means to disable their function for the inspection of the metering system.

6.0 Revision

The original bulletin was issued on September 1, 2008 and became effective on October 1, 2008.

6.1 The purpose of revision 1 was to allow the required information on a printed ticket to be stated for the blended product only where the individual component information can be provided upon request. In this revision, the scope was also changed to include blend dispensers and a new section was added, "Provisions for Testing", which requires the design of such systems to allow for the testing of the individual meters with the products they measure at their normal flow rates, Minor editorial corrections were also made.

6.2 The purpose of revision 2 is to clarify section 5.4 so that the trade meter in a side stream blending system can be tested with either the pure product or a blend that it normally measures. This allows the system to be tested with pure product if the product is returned to storage or with a blend, if it is not.

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